MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY 11747210

TARGET COINCIDENCE BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a target coincidence printed circuit board, a part of Range Computer Test Panel, Subassembly MX-8049/TPM-22, a part of Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F11747212	Schematic, Target Coincidence Board

FSC 4931

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F11747210	Printed Wiring Board Assembly, Target Coincidence Board
F10554938	Connector
D10548169	Integrated Circuit

PACKAGING DATA SHEET

F11747210	Packaging of Target Coincidence Board

(Copies of specification, standards, drawings, and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The Target Coincidence Board shall be manufactured in accordance with Drawing F11747210 and drawings pertaining thereto.

3.2 General specification.- Where applicable, the target Coincidence Board shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence.

b. Dimensions and tolerances.

c. Inorganic protective surface finishings.

d. Part identification and marking.

e. Workmanship.

f. Soldering electrical connections.

3.2.2 MIL-C-46357.

a. Etched circuit assemblies.

b. Wiring.

c. Threaded fasteners.

d. Marking.

e. Finishes.

3.3 Environmental service conditions.

3.3.1 Vibration.- The target coincidence board shall be mounted on the vibration table in normal service position. The target coincidence board shall meet the requirements of 3.4 and 3.5 after being

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subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

FREQUENCY RANGE	DOUBLE AMPLITUDE
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock.- The target coincidence board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 $\pm$ 1 milliseconds. The G value of each impulse shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating.- The target coincidence board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating.- The target coincidence board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating.- The target coincidence board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage).- The target coincidence board shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity-nonoperating.- The target coincidence board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity.- Each conductor shall be continuous electrically as specified on the schematic diagram F11747212.

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3.4.2 Resistance.- Resistance shall be 100 ohms ± 10% between the following terminals.

FROM	TO
P1-1	P1-2
P1-5	P1-7
P1-20	P1-22

3.5 Performance.

3.5.1 Conditions of operation.- Unless otherwise specified, the requirements of 3.5.2 shall be met with the circuit connections shown in Figure 1. Signal input requirements to the circuit of Figure 1 shall comply to the signal requirements given in Table I.

TABLE I - SIGNAL INPUTS

ITEM	SIGNAL INPUT	CHARACTERISTICS	APPLICATION POINTS
1	Mod TO Trigger	Positive Pulse from OV to not less than 15 Vdc. 0.2 to 0.6 usec Pulse Width @ 50% Points. Rise time of 40 nsec max into 50 ohms as measured between 10% & 90% points. 20 KHz ± 200 Hz PRF.	Test Fixture J6
2	E/L Trigger	Positive Pulse from OV To 4.0 ± 0.5 Vdc. Pulse width manually variable between 5 usec and 20 usec. Falling edge occurring at mod TO. PRF=20 KHz ± 200 Hz.	Signal - J1, Terminal 20 Return - J1, Terminal 22
3	Target Trigger	14 ± 1 usec positive pulse OV to 4.0 ± 0.5 Vdc Rising edge occurs at Mod TO	Signal - J1, Terminal 51 Return - J1, Terminal 24
4	Doppler	1 KHz ± 10 Hz Square Wave OV to 4.0 ± 0.5 Vdc	Signal - J1, Terminal 35 Return - J1, Terminal 24
5	2560M	50% Square Wave OV to 4.0 ± 0.5 Vdc Period = 16.38 ± 0.2 Sec	Signal - J1, Terminal 43 Return - J1, Terminal 24

3.5.2 Output signals.

3.5.2.1 Sum video.- With the conditions of 3.5.1 applied, the signal present at TP1 on the target coincidence board shall be a positive pulse from OV to 4.0 ± 1.0 Vdc. The pulse width shall be continuously

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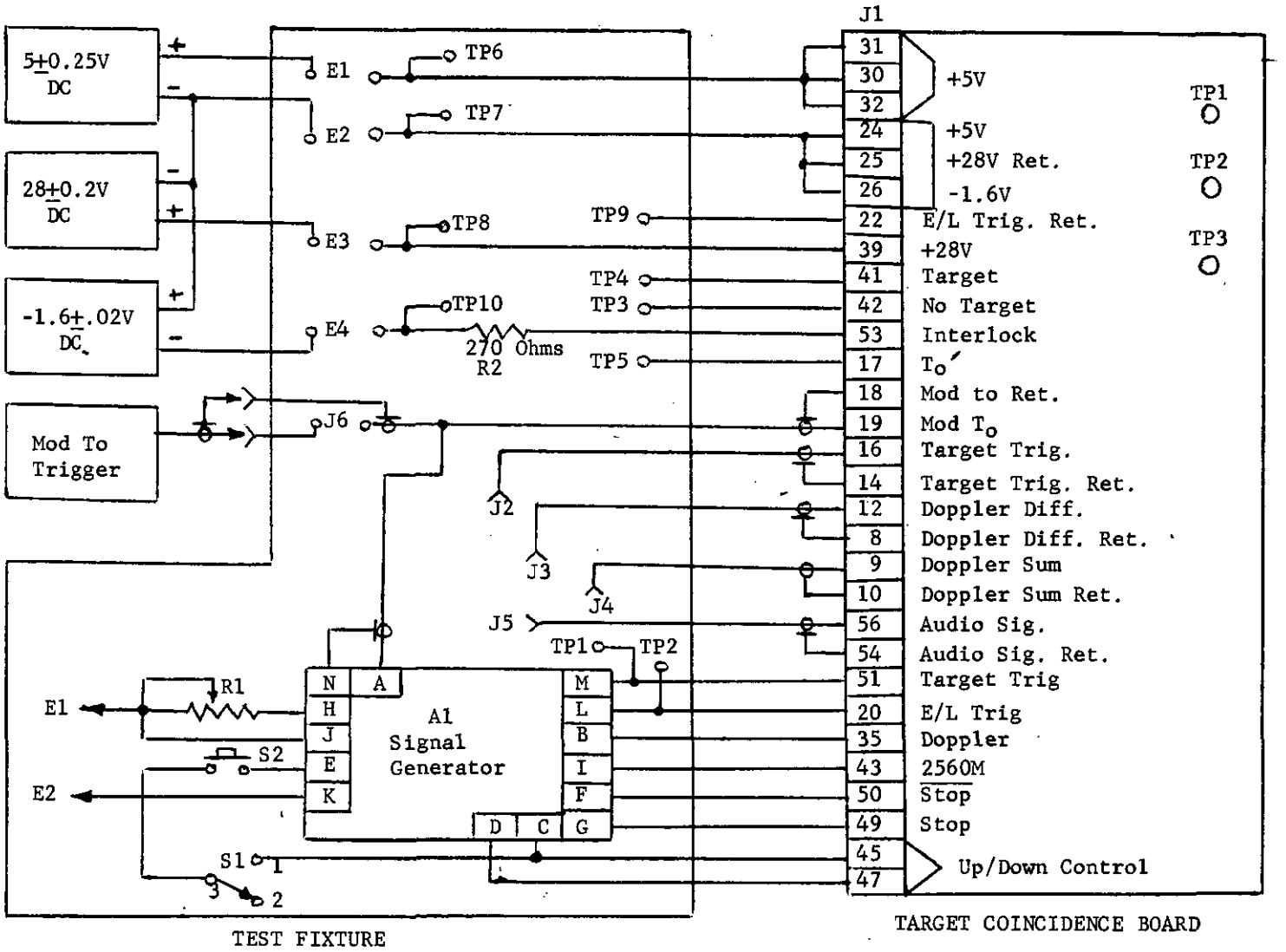


FIGURE 1 - CIRCUIT CONNECTIONS

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variable between 0.3 usec and 0.6 usec with variation of R4 located on the target coincidence board. The rising edge of the positive pulse shall be coincident with the falling edge of the target trigger pulse (item 3, Table I).

3.5.2.2 Difference video.- With the conditions of 3.5.1 applied, the signal present at TP2 on the target coincidence board shall be a positive pulse from)V to 4.0 ± 1.0 Vdc. The pulse width shall be continuously variable between 0.3 usec and 0.6 usec with variations of R5 located on the target coincidence panel. The rising edge of the positive pulse shall be coincident with the falling edge of the sum video pulse (3.5.2.1).

3.5.2.3 E/L video.- With the conditions of 3.5.1 applied, the signal present at TP3 on the target coincidence board shall be a positive pulse from OV to 4.0 ± 1.0 Vdc. The pulse width shall be continuously variable between 0.3 usec and 1.2 usec with variations of R6. The rising edge of the positive pulse shall be coincident with the rising edge of the E/L trigger pulse (Item 2, Table I).

3.5.2.4 Doppler sum.- With the conditions of 3.5.1 applied, the signal present at J4 on the test fixture shall be a 1 KHz ± 100 Hz square wave, from OV to not less than +2 Vdc. This signal shall be present when the E/L video pulse (3.5.2.3) is coincident with either the Sum video (3.5.2.1) or Difference video (3.5.2.2) pulse. When there is no coincidence, there shall be a dc level of -.86 ± .04 Vdc at J4.

3.5.2.5 Doppler difference.- With the conditions of 3.5.1 applied, the signal present at J3 on the test fixture shall be a 1 KHz ± 100 Hz square wave, from OV to not less than +2 Vdc. This signal shall be in phase with the DOPPLER SUM (3.5.2.4) when SUM VIDEO only (3.5.2.1) is at least 25 ± 2 nsec coincident with E/L Video (3.5.2.3); and 180° out of phase with the SUM DOPPLER (3.5.2.4) when DIFFERENCE VIDEO only (3.5.2.2) is at least 25 ± 2 ns coincident with E/L VIDEO (3.5.2.3). When there is no coincidence, there shall be a dc level of -0.86 ± .04 Vdc.

3.5.2.6 Target Trigger.- With the conditions of 3.5.1 applied the signal present on J2 on the test fixture shall be a positive pulse from OV to not less than 15 Vdc. The pulse width shall be the same pulse width as the SUM VIDEO (3.5.2.1) pulse, ± 20 ns, and the same PRF.

3.5.2.7 Audio.- With the conditions of 3.5.1 applied, the signal present at J5 on the test fixture shall be a 1 KHz ± 10 Hz square wave, OV to 4.0 ± 1.0 Vdc. This signal shall be in phase with the DOPPLER (Item 4, Table I) input signal.

3.5.2.8 No target ind.- With 4 ± 0.5 Vdc applied to terminal 49 on J1 and 0 ± 0.5 Vdc applied to terminal 50, the resistance measured between TP3 on test fixture and 24 on J1 shall be less than 2.5 ohms.

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3.5.2.9 Target ind.- With OV, dc applied to terminal 49 on J1 and 4 ± 0.5 Vdc applied to terminal 50, the resistance measured between TP4 on test fixture and 24 on J1 shall be less than 2.5 ohms.

3.5.2.10 One second fade.- When the TARGET TRIGGER (Item 3, Table I) and 2560M (Item 5, Table I) signals are applied, the SUM VIDEO (3.5.2.1) measured on TP1 shall fade for 1 ± 0.2 sec. The fade shall occur 8.2 ± 0.2 sec after 4 ± 0.5 Vdc and 0 Vdc are simultaneously applied to terminal 49 and terminal 50 respectively. The fade shall occur in either position of switch S1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or Contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three Target Coincidence Boards manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein, and test methods and procedures in 4.6. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

100% INSPECTION

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Continuity	3.4.1	4.6.2.1
102.	Resistance	3.4.2	4.6.2.2
103.	Conditions of Operation	3.5.1	4.6.3.1
104.	Sum Video	3.5.2.1	4.6.3.2
105.	Difference Video	3.5.2.2	4.6.3.3
106.	E/L Video	3.5.2.3	4.6.3.4
107.	Sum Doppler	3.5.2.4	4.6.3.5
108.	Doppler Difference	3.5.2.5	4.6.3.6
109.	Target Trigger	3.5.2.6	4.6.3.7
110.	Audio	3.5.2.7	4.6.3.8
111.	No Target Ind	3.5.2.8	4.6.3.9
112.	Target Ind	3.5.2.9	4.6.3.10
113.	One Second Fade	3.5.2.10	4.6.3.11

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One target coincidence board out of the first 25 produced and one out of each 25 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

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TABLE III - SPECIAL SAMPLING

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specification	3.2	Visual-Tactile
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3
306.	Altitude - nonoperating	3.3.3.2	4.6.1.3
307.	Temperature	3.3.4	4.6.1.4
308.	Humidity - nonoperating	3.3.5	4.6.1.5

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

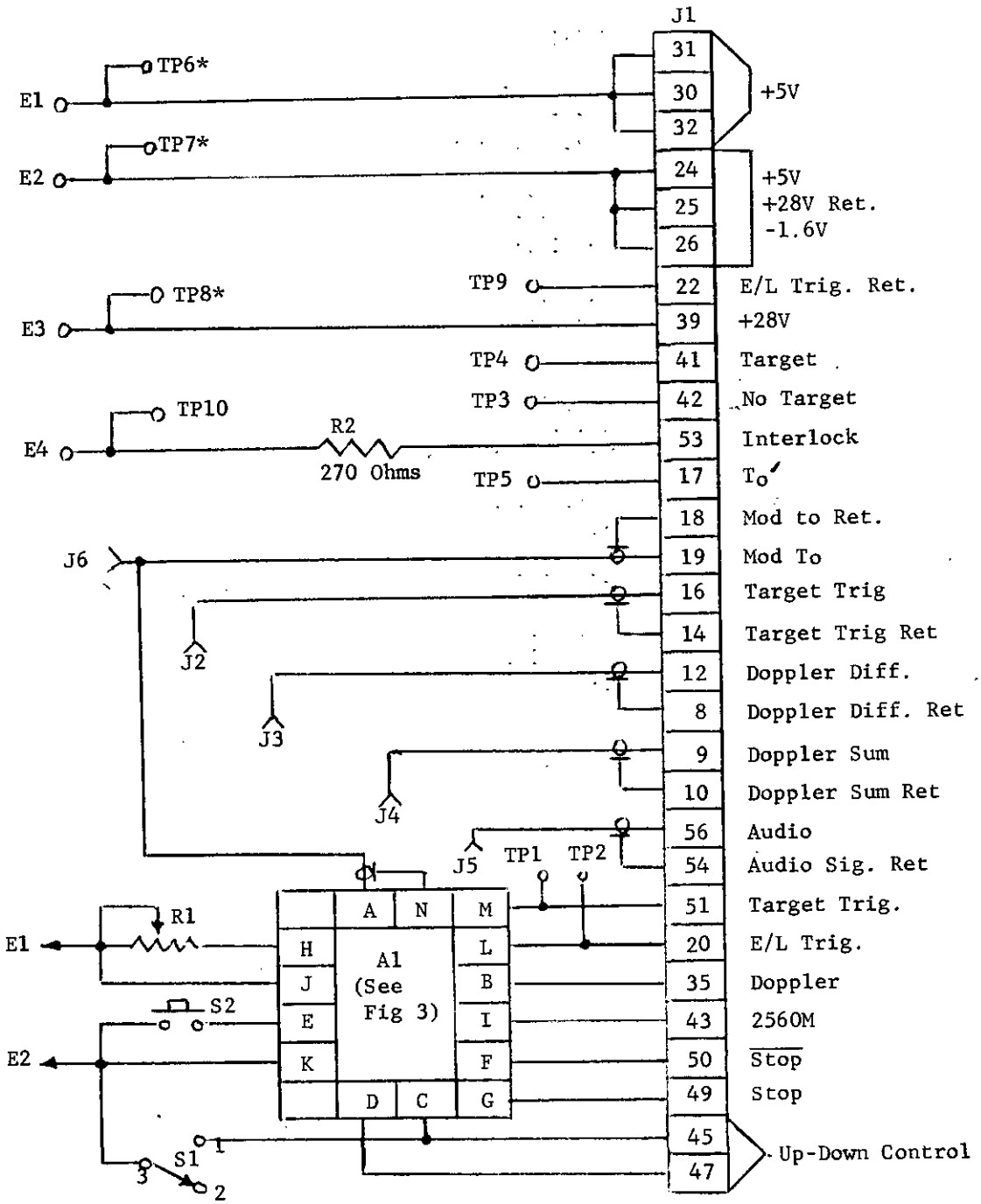
4.5.2.2 Test equipment.- In conjunction with 4.5.2.1, the test equipment found in Table IV shall be used in the performance of the applicable tests as specified in 4.6.

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TABLE IV - TEST EQUIPMENT LIST

ITEM	EQUIPMENT	DESCRIPTION
1.	Power Supply	5.0 ± 0.25 Vdc, Regulated AC ripple: Less than 50 MV peak-to-peak Capacity: 600 ma
2.	Power Supply	28 ± 0.2 Vdc, Regulated AC ripple: Less than 50 mv peak-to-peak Capacity: 200 ma
3.	Signal Generator	300 ± 10 ns wide pulse, repeition rate 20 KHz ± 100 Hz, 10 ± .1V peak-to-peak into 50 ohms.
4.	Oscilloscope	Dual channel; rise time of scope plus probe not greater than 10 nsec, sweep period less than 50 nsec Input resistance: 8 megohms or greater
5.	Test fixture	As specified in Figure 2. See Table V for test fixture parts list.
6.	Ohmmeter	Accuracy and resolution of ±10%, to measure 200 Ohms.
7.	Power Supply	0 to +5V adj., regulated AC ripple: Less than 50 mv peak-to-peak Capacity: 400 ma
8.	Voltmeter	Accuracy and resolution of ±1% 0 to 10 Volts, Digital.

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*Connect TP6 thru TP8 as close to J1 as possible.

FIGURE 2 - TEST FIXTURE

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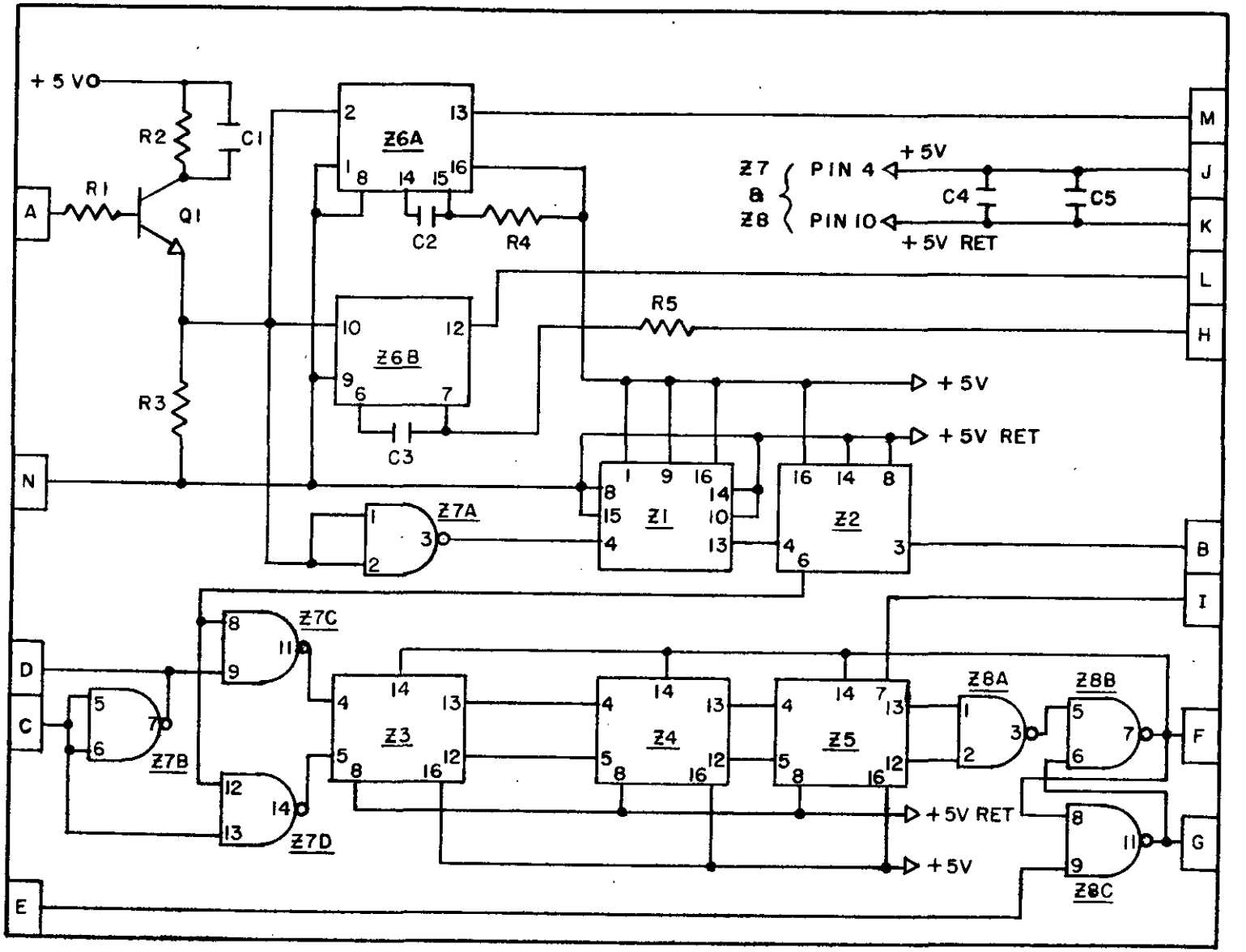


FIGURE 3 – TEST FIXTURE SIGNAL GENERATOR AI

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TABLE V - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
A1	Test fixture signal generator in accordance with Figure 3. See Table VI for parts list.
E1, E3, E4	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector, per Frankford Arsenal F10554938 or equivalent.
J2 thru J6	Connector, BNC Receptacle
TP1 thru TP 10	Tip Jack, H. H. Smith 1501 or equivalent.
S1	Switch SPDT.
S2	Momentary Switch SPST.
R1	Variable Resistor, 10K Ohms > 1/4 watt, 10 turn.
R2	Resistor, 270 Ohms ± 1%, 5 watt.

TABLE VI - TEST FIXTURE SIGNAL GENERATOR PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
C1, C4, C5	Capacitor 0.1 uF ± 10%, @ 50 Vdc
C2, C3	Capacitor 4700 pF ± 10%, @ 50 Vdc
R1, R4	Resistor, 10K Ohms ± 10%, 1/4 watt
R2, R3	Resistor, 100 ohms ± 10%, 1/4 watt
Z1, Z2, Z3, Z4, Z5	Integrated Circuit Texas Instruments P/N SN54193
Z6	Integrated Circuit Texas Instruments P/N SN54123J
Z7, Z8	Integrated Circuit FA P/N 10548169
R5	Resistor, 3K Ohms ± 10%, 1/4 watt

4.6 Test methods and procedures.

4.6.1 Environmental.- Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

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4.6.1.1 Vibration.- The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure criteria of 4.9.1.3 does not apply. The frequency and amplitude limits of paragraph 3.3.1 of this specification shall be used. After vibration, the target Coincidence Board shall be tested as given in the subparagraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock.- This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 g's. After the shock impulses, the target Coincidence Board shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - Non-operating and operating.- This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the target Coincidence Board shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the target Coincidence Board shall be tested in accordance with 4.6.2.2 and 4.6.3.

4.6.1.5 Humidity-non-operating.- This test shall be performed in accordance with Procedure I, Method 507 of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical.

4.6.2.1 Continuity.- Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.6.2.2 Resistance.- Resistance shall be measured with an ohmmeter, item 6 of Table IV for compliance with 3.4.2.

4.6.3 Performance.

4.6.3.1 Conditions of Operation.- Unless otherwise specified, all subsequent tests will be performed with the D.C. power supply connections and signal connections as follows: Connect up the power supplies, items 1, 2, and 7 of Table IV; the signal generator, item 3 of Table IV; and the test fixture, item 5 of Table IV; to conform to the schematic of Figure 1. Adjust the D.C. power supplies to the values indicated. Apply

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the MOD TO trigger (item 1, Table I) to test fixture J6.

4.6.3.2 Sum Video.- Connect Channel 1 of the Oscilloscope, item 4 of Table IV, to TP1 on the target Coincidence Board. Check for the presence of the positive pulse per 3.5.2.1. Vary R4 on the target Coincidence Board and check the pulse width for conformance to 3.5.2.1 requirements. Connect channel 2 of the Oscilloscope to TP1 of the test fixture (+) and E2 of the test fixture (-), and check the phase requirements of 3.5.2.1.

4.6.3.3 Difference Video.- Connect channel 1 of the oscilloscope, item 4 of Table IV, to TP2 on the target Coincidence Board. Check for the presence of the positive pulse per 3.5.2.2. Vary R5 on the target Coincidence Board and check the pulse width for conformance to 3.5.2.2 requirements. Connect channel 2 of the oscilloscope to TP1 on the target Coincidence Board, and check the phase requirements of 3.5.2.2.

4.6.3.4 E/L Video.- Connect channel 1 of the oscilloscope, item 4, of Table IV, to TP3 on the target Coincidence Board. Check for the presence of the positive pulse per 3.5.2.3. Vary R6 on the target Coincidence Board and check the pulse width for conformance to 3.5.2.3. Connect channel 2 of the oscilloscope to TP2 of the test fixture (+) and TP9 of the test fixture (-), and check for the phase requirements of 3.5.2.3.

4.6.3.5 Doppler Sum.- Connect channel 1 of the oscilloscope, item 4, of Table IV, to J4 of the test fixture. Check for the presence of the 1 KHz square wave defined in 3.5.2.4. If the square wave is present disconnect the oscilloscope. Connect channel 1 of the oscilloscope to TP3 on the target Coincidence Board, and channel 2 alternately to TP1 and TP2. Check for coincidence of the E/L video pulse, with either the SUM video or difference video pulses per the requirements of 3.5.2.4. If the square wave is not present at terminal J4 check for the dc level per 3.5.2.4 requirements.

4.6.3.6 Doppler difference.- Connect channel 1 of the oscilloscope, item 4 of Table IV, to J3 on the test fixture. Check for the presence of the 1 KHz square wave per the requirements of 3.5.2.5.

4.6.3.6.1 If the signal is not present go to 4.6.3.6.2. If the signal is present, connect channel 2 of the oscilloscope to J4 of the test fixture. Check to see if channel 2 signal is in-phase, or out-of-phase with channel 1 signal. If signals are in phase, connect channel 1 to TP1, and channel 2 to TP3. Determine that the two displayed signals are at least 25 $\pm$ 2 ns coincident per the requirements of 3.5.2.5. If signals are out-of-phase, connect channel 1 to TP2 and channel 2 to TP3. Determine that the two displayed signals are at least 25 $\pm$ 2 ns coincident per the requirements of 3.5.2.5.

4.6.3.6.2 When the signal is not present, check for a dc level of -0.86 $\pm$ .04 per the requirement of 3.5.2.5.

4.6.3.7 Target trigger.- Connect channel 1 of the oscilloscope,

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item 4 of Table IV, to J2 on the test fixture. Check for the presence of the positive pulse per the requirements of 3.5.2.6. Connect channel 2 of the oscilloscope to TP1 on the target coincidence board. Check for pulse width and PRF per the requirements of 3.5.2.6.

4.6.3.8 Audio.- Connect channel 1 of the oscilloscope, item 4 of Table IV, to J5 on the test fixture. Check for the presence of the 1 KHz square wave per the requirements of 3.5.2.7. Connect channel 2 to the application points of the Doppler signal given in Table I, as item 4. Check for the phase requirements of 3.5.2.7.

4.6.3.9 No target ind.- Connect a voltmeter, item 8 of Table IV, to terminal 49 of J1(+) and return to terminal 50(-). Check for the presence of 4 $\pm$ 0.5 Vdc per the requirements of 3.5.2.8. If the dc signal is not present momentarily depress switch S2. When the dc signal is present proceed to measure the resistance between TP3 on test fixture and 24 of J1 using an ohmmeter, item 6 of Table IV. Determine that the resistance measured meets the requirements of 3.5.2.8.

4.6.3.10 Target Ind.- Connect a voltmeter, item 8 of Table IV, to terminal 49 of J1(+) and return to terminal 50(-). Check for the presence of 0 $\pm$ 0.2 Vdc per the requirements of 3.5.2.9. If the dc signal is not present momentarily depress switch S2. When the dc signal is present proceed to measure the resistance between TP4 on Test Fixture and 24 of J1 using an ohmmeter, item 6 of Table IV. Determine that the resistance measured meets the requirements of 3.5.2.9.

4.6.3.11 One second fade.- The steps given below shall be followed in sequence.

a. Connect a voltmeter, item 8 of Table IV, to terminal 49 of J1(+) and return to terminal 50(-). Check for the presence of 0 $\pm$ 0.2 Vdc. If the dc signal is not present momentarily depress switch S2 on the target coincidence board. When the dc signal is present disconnect the voltmeter.

b. Check for the presence of the TARGET TRIGGER (item 3, Table I) signal, and the 2560M (item 5, Table I) signal, by using the oscilloscope, item 4 of Table IV, on the application points for these signals listed in Table I.

c. Connect channel 1 of the oscilloscope, to TP1 on the target coincidence board.

d. Monitor signal on oscilloscope and depress switch S2 momentarily. Confirm the presence of signal fade per the requirements of 3.5.2.10. Change the position of S1 on the test fixture and repeat steps a through d.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11747210. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use.- This Circuit Card Assembly is a part of Range Computer Test Panel, Subassembly MX-8049/TPM-22 which is used for troubleshooting and performance checking of specific weapon system components.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Block of serial numbers to identify each target coincidence board.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 4931-A132

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-48581(MU) Circuit Card Assembly 11747210 Target Coincidence Board

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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